

CELLGUARD · DESIGN DECISION RECORD

CellGuard recommendations accepted

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DOCUMENT CONTROL

Document ID	CGD-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellguard		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items left open on 2026-09-25 decided by Amish as recommended (CGD-DEC-001)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. The items in Table 2, which had no recommendation on 2026-09-25, were decided by Amish on 2026-10-02 as later recommended: "i approve your recommendations for all 555 open decisions." (CGD-DEC-001).

■ Context

CGD-DDR-001 v0.1 recorded the TRL 2 review items as adopted for TRL 3 work, open for Amish's review, and left the budget, the firmware licensing rule and the TRL 3 engineering proposals as proposed, awaiting Amish. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that carried a recommendation is therefore decided as recommended. Where a recommendation offered several options, the recommended option is the decision. Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, and the project stays at TRL 3 with a TRL 3 target.

■ Options considered

The options for each item are in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions), CGD-DDR-001 and CGD-CAL-001 section 12. They are not repeated here.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

# (DDR-001)	Item	Decision	What changed in the repo
1	Front end	TI BQ76952 class	Wording only: CGD-PRC-001 v0.4, <code>bom/bom.csv</code> line 3 note
2	Microcontroller and firmware base	STM32G0B1 class building on the Libre Solar BMS firmware	Wording only: CGD-PRC-001 v0.4, <code>bom/bom.csv</code> line 6 note
3	Secondary protector	BQ77216-class protector with SCP fuse	Wording only: CGD-PRC-001 v0.4, <code>bom/bom.csv</code> line 14 note
4 and 11	Budget	Option (a): raise <code>budget_usd</code> from \$120 to \$140	<code>project.yaml</code> <code>budget_usd: 140</code> ; CGD-REQ-001 v0.4 R16 target \$140; CGD-CAL-001 v0.2 section 11 and <code>sizing.py</code> judge cost against \$140 (\$134.00, 95.7 %, \$6.00 margin); README budget line; R16 moves from not met to met
5	Current rating	40 A continuous	Wording only: CGD-PRB-001 v0.4, CGD-PRC-001 v0.4
6	Switching side	High-side N-channel switching	Wording only: CGD-PRC-001 v0.4
7	Chemistry scope	LFP and NMC-capable hardware, LFP firmware first	Wording only; still needs the SwapCell project's agreement (cross-repo action in <code>docs/REVIEW.md</code>)
8	CAN message set	SwapCell interface v0.3 message set as a CellGuard profile	Wording only; still needs the SwapCell project's agreement (cross-repo action)
10	Apache-2.0 firmware in an MIT repo	Reused files keep Apache-2.0 and their notices in <code>firmware/third_party/</code> and are listed in <code>LICENSE-SOFTWARE</code>	Note added to <code>LICENSE-SOFTWARE</code> and to the README licenses section; CGD-PRC-001 v0.4 choice 3; CGD-REQ-001 v0.4 R15. No firmware is written (TRL 4, on hold)
14a	Protection settings	SCD 100 mV at 15 μ s, OCD2 40 mV at 20 ms, OCD1 24 mV at 320 ms, OCC 12 mV	CGD-CAL-001 v0.2 Table 2 marked decided; CGD-REQ-001 v0.4 R3 cites them as defaults
14b	MOSFET package	Top-side cooled TOLT package	<code>bom/bom.csv</code> line 4 spec (TOLL no longer listed; price unchanged at \$2.50); CGD-PRC-001 v0.4 Table 1; README key components. The model already represents the packages as blocks of the same size, so the geometry, STEP, STL and drawing CGD-DWG-001 are unchanged (stays Rev P1)
14c	Precharge switch	DPAK-class P-channel MOSFET	<code>bom/bom.csv</code> line 7 note; CGD-PRC-001 v0.4 Table 1 (already the specified part)
14d	Current calibration	One-point gain calibration at build	CGD-REQ-001 v0.4 R9; CGD-CAL-001 v0.2 section 6. Decided but on hold: the build and calibration are TRL 4 work
15 (R16 part)	Response to R16	The recommended \$140 budget	Covered by item 4

■ Items left open on 2026-09-25

Table 2. Items without a recommendation on 2026-09-25, decided by Amish on 2026-10-02.

# (DDR-001)	Item	Status
9	First test partner (solar installer, e-bike repair shop or makerspace)	Decided by Amish, 2026-10-02, as recommended: a small off-grid solar installer that builds 16S LFP packs is the first candidate to approach, testing on their bench rather than in a customer system (CGD-DEC-001)
12	Cover material: printed flame-retardant polycarbonate or sheet aluminium	Decided by Amish, 2026-10-02, as recommended: printed flame-retardant polycarbonate, as drawn; sheet aluminium is not used (CGD-DEC-001)
13	Low-cost method to check the state-of-charge estimate	Decided by Amish, 2026-10-02, as recommended: cycle a 4S 20 Ah LFP pack through seven days of partial cycles with a bench supply and a low-cost DC electronic load, count charge with a calibrated reference shunt and meter, then discharge fully to measure the true state of charge (CGD-DEC-001)
15 (R6)	Board loss 4.44 W against 4 W: 1.5 mΩ-class MOSFETs or the SCP fuse moved off the board	Decided by Amish, 2026-10-02, as recommended: 1.5 mΩ-class switches in the TOLT package, about 3.5 W (CGD-DEC-001)
15 (R8)	State of charge on 20 Ah packs: restate R8 for 50 Ah and larger, a full-charge prompt, or a 0.5 mΩ shunt	Decided by Amish, 2026-10-02, as recommended: a firmware prompt for a full charge when about five days pass without one, with R8's partial-cycling clause restated to apply between prompted full charges; no 0.5 mΩ shunt (CGD-DEC-001)
15 (R14)	Mass 0.646 kg against 0.6 kg: a 3 mm plate (0.581 kg) or relax the target to 0.65 kg	Decided by Amish, 2026-10-02, as recommended: R14 relaxed to 0.7 kg, 4 mm plate kept (CGD-DEC-001)

■ Consequences

- Requirement status on paper moves from 9 met, 4 not met and 3 at risk to 10 met, 3 not met (R6, R8, R14) and 3 at risk (R2, R4, R7).
- `budget_usd` is \$140. The \$134.00 BOM leaves a \$6.00 margin, which the unconfirmed SCP fuse price (line 14) could use up.
- Specifying the TOLT package fixes the thermal result at 53.7 °C junction at 40 A; the TOLL fallback figure (61.4 °C, 97.7 °C bound at 80 A for 10 s) is kept in CGD-CAL-001 for reference only.
- Items 7 and 8 still depend on the SwapCell project, and MotionCore must run its CAN bus at 250 kbit/s to share the profile. These are listed as cross-repo actions in `docs/REVIEW.md`; no other repo was edited.
- The one-point calibration and all firmware are TRL 4 work and remain on hold by Amish's instruction. `trl` and `trl_target` stay at 3.